

ECON403/503

Preliminary Material: Axiomatic Foundations of Probability and Random Variables & Continuous RV's and Jointly Distributed RVs

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Basic Ingredients: Intuitive Definitions

Outcomes

The mutually exclusive potential results of a random process are called outcomes. For example, if you shoot a penalty kick in football (soccer) possible outcomes are: i) you score a goal, ii) you do not score a goal.

Sample Space

The set of all possible outcomes is called the sample space.

Event

An event is a subset of the sample space, that is, an event is a set containing one or more outcomes.

Probability of an Outcome

For each outcome, we associate a number between zero and one that will be called the probability of the outcome.

Probability of an Event

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Definition 1

The set Ω , of all possible outcomes of a particular experiment is called the **sample space** for the experiment. An **outcome** of the sample space is denoted with ω .

Example 2

- Tossing a coin: the sample space is $\Omega = \{H, T\}$.
- Observing the TOEFL scores of randomly selected students: $\Omega = \{0, 1, 2, \dots, 120\}$.
- Observing the income of randomly selected individuals: $\Omega = [0, \infty)$.

Definition 3

An **event** is any collection of possible outcomes of an experiment, that is, any subset of Ω (including Ω itself).

Creating new events from given ones

Union: $A \cup B = \{x \in \Omega : x \in A \text{ or } x \in B\}$.

Intersection: $A \cap B = \{x \in \Omega : x \in A \text{ and } x \in B\}$.

Complementation: $A^c = \{x \in \Omega : x \notin A\}$. We can also take the complement of A in B :
 $B \setminus A = \{x \in \Omega : x \in B \text{ and } x \notin A\}$.

The operations of union and intersection can be extended to countable or infinite collection of sets (with the same notation as the one we use for sums $\sum_i$ and products $\prod_i$.)

Definition 4

Two events A and B are **disjoint** if $A \cap B = \emptyset$. The events $A_1, A_2, \dots$ are **pairwise disjoint** if $A_i \cap A_j = \emptyset$ for all $i \neq j$.

Definition 5

If $A_1, A_2, \dots$ are pairwise disjoint and $\bigcup_{i=1}^{\infty} A_i = \Omega$, then the collection $A_1, A_2, \dots$ forms a **partition** of Ω .

- For each event A in the sample space we want to associate with A a number between zero and one that will be called the probability of A , denoted $\mathbb{P}(A)$.
- Although it may seem natural to define the domain of $\mathbb{P}$ as all subsets of Ω , unfortunately, matters are not that simple.
- There are some technical difficulties to overcome, but we will not dwell too much on these. However, a firm understanding of the concepts that we are going to discuss later in the course requires at least some passing familiarity with the following.

Definition 6

A collection of subsets of Ω is called a **sigma algebra**, denoted by $\mathcal{B}$, if it satisfies the following three properties:

- 1 $\emptyset \in \mathcal{B}$ (the empty set is an element of $\mathcal{B}$).
- 2 If $A \in \mathcal{B}$ then $A^c \in \mathcal{B}$ ($\mathcal{B}$ is closed under complementations).
- 3 If $A_1, A_2, \dots \in \mathcal{B}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$ ($\mathcal{B}$ is closed under countable unions).

Remark:

Associated with any sample space Ω we can have many different sigma algebras. For example, the collection $\{\emptyset, \Omega\}$ is a sigma algebra. We will usually be concerned only with the smallest sigma algebra that contains all of the open sets in a given sample space Ω .

Example 7

- If Ω is finite or countable, then the technical difficulties do not arise, and we can define for a given sample space Ω

$$\mathcal{B} = \{\text{all subsets of } \Omega, \text{ including } \Omega \text{ itself}\}.$$

- If Ω is uncountable, it is not an easy task to describe $\mathcal{B}$. In this case, $\mathcal{B}$ is chosen to contain any set of interest.
- Suppose $\Omega = (-\infty, \infty)$. $\mathcal{B}$ is chosen to contain all sets of the form

$$[a, b], \quad (a, b], \quad (a, b), \quad \text{and} \quad [a, b)$$

The Kolmogorov Axioms

Given a sample space Ω and an associated sigma algebra $\mathcal{B}$, a **probability function** is a function $\mathbb{P}$ with domain $\mathcal{B}$ and range $[0, 1]$, that is, $\mathbb{P} : \mathcal{B} \rightarrow [0, 1]$ that satisfies:

- 1 $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{B}$.
- 2 $\mathbb{P}(\Omega) = 1$.
- 3 If $A_1, A_2, \dots \in \mathcal{B}$ are pairwise disjoint, then $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

The previous are referred to as the **Axioms of Probability** or the *Kolmogorov Axioms*.

Definition 8

The triple $(\Omega, \mathcal{B}, \mathbb{P})$, where Ω is the sample space, $\mathcal{B}$ is a sigma-algebra over Ω and $\mathbb{P} : \mathcal{B} \rightarrow [0, 1]$ is a probability function satisfying the axioms of probability, is called a **probability space**.

Theorem 9

If $\mathbb{P}$ is a probability function and A and B are any sets in $\mathcal{B}$, then

- ① $\mathbb{P}(\emptyset) = 0$.
- ② $\mathbb{P}(A) \leq 1$.
- ③ $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- ④ $\mathbb{P}(B \cap A^c) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- ⑤ $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- ⑥ If $A \subset B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Proof.

Homework!



Example 10

Experiment: Flip a coin four times. Assume the coin is fair.

- The sample space Ω contains the following outcomes: $HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT, THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH$, and $TTTT$
- We can take the sigma-algebra $\mathcal{B}$ to be the set of all subsets of Ω . This collection of sets would be very cumbersome to describe since it has 2^{16} subsets. Nevertheless, it contains all possible events that we can possibly describe.
- For any outcome $\omega \in \Omega$ in this example, we define $\mathbb{P}(\omega) = 1/16$. Note that this does not follow from the Axioms of Probability but it is outside of the axioms. We have used a symmetric interpretation of probability (or intuition) to impose this requirement: if we have a fair coin, then all outcomes must be equally likely. Note that this assignment satisfies the axioms.
- However, all other assignments of $\mathbb{P}$ could have been valid as long as they satisfy the axioms. For example, if we assume the coin is not fair, we could have assigned other values.
- For any event $A \in \mathcal{B}$, define $\mathbb{P}(A) = \sum_{\{\omega_i \in A\}} \mathbb{P}(\omega_i)$.
- Thus, the triple $(\Omega, \mathcal{B}, \mathbb{P})$ describes formally the experiment of flipping a fair coin four times.

- In many experiments it is easier to deal with a summary variable than with the original probability structure.
- From the previous example, if we are only interested in the number of tails that appear after we complete the four coin tosses, then we might as well define a variable that counts the number of tails in the experiment. Say
 X = number of tails in the four coin tosses. X can take on five values: $\{0, 1, 2, 3, 4\}$.
- In defining the quantity X , we have defined a mapping (a function) from the original sample space, to a new sample space, usually a set of real numbers.

Definition 11

A **random variable** is a function from a sample space Ω into the real numbers, that is, $X : \Omega \rightarrow \mathbb{R}$. A **realization** of the random variable is denoted as $X(\omega) = x$, where $\omega \in \Omega$ and $x \in \mathbb{R}$.

- In defining a random variable, we have defined a new sample space (the range of the random variable.)
- We must make sure that our original probability function can be used for the random variable.
- I will present here how to do this. For the sake of simplicity, assume that Ω is finite:
 $\Omega = \{\omega_1, \dots, \omega_n\}$. Let $\mathcal{B}$ be the sigma-algebra containing *all* subsets of Ω , and let $\mathbb{P} : \mathcal{B} \rightarrow [0, 1]$ be a probability function.
- Define a random variable X with range $\mathcal{X} = \{x_1, \dots, x_m\}$. We can define a probability function P_X in the following way. We will observe $X = x_i$ if and only if the outcome of the random experiment is an $\omega_j \in \Omega$ such that $X(\omega_j) = x_i$. Thus,

$$P_X(X = x_i) = \mathbb{P}(\{\omega_j \in \Omega : X(\omega_j) = x_i\}).$$

- Note that $\{\omega_j \in \Omega : X(\omega_j) = x_i\}$ is an event that belongs to $\mathcal{B}$. Thus the right-hand side of the expression above is well defined.
- The definition above makes P_X a valid probability function for the random variable X .
- P_X is an **induced** probability function on $\mathcal{X}$.

- The construction in the last slide generalizes to more general probability spaces, not necessarily finite.
- If $(\Omega, \mathcal{B}, \mathbb{P})$ is a probability space, and X is a random variable, let $X^{-1}(x) = \{\omega \in \Omega : X(\omega) = x\}$ denote the **inverse image** of X . With a slight abuse of notation, if $A \subset \mathcal{X}$ where $\mathcal{X}$ is the range of X , then $X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\}$.
- If $\mathcal{B}(\mathcal{X})$ denotes a sigma-algebra over the range of $\mathcal{X}$, then for any $A \in \mathcal{B}(\mathcal{X})$:

$$P_X(X \in A) = \mathbb{P}(X^{-1}(A))$$

- Then $(\mathcal{X}, \mathcal{B}(\mathcal{X}), P_X)$ becomes our **induced probability space**. This is the usual starting point of the analysis, instead of the base probability space.

Example 12

Experiment: Flip a fair coin four times, continued. Suppose that X counts the number of tails:

- $\mathcal{X} = \{0, 1, 2, 3, 4\}$.
- We can take $\mathcal{B}(\mathcal{X})$ as the set of all subsets of $\mathcal{X}$. This collection has $2^5 = 32$ subsets, so I will write it up.
- Let's look at the inverse images:
 - $X^{-1}(0) = \{HHHH\}$
 - $X^{-1}(1) = \{HHHT, HHTH, HTHH, THHH\}$
 - $X^{-1}(2) = \{HTTH, TTHH, HHTT, THTH, HTHT, THHT\}$
 - $X^{-1}(3) = \{HTTT, THTT, TTHT, TTTH\}$
 - $X^{-1}(4) = \{TTTT\}$
- Therefore, the respective probabilities are
 - $P_X(X = 0) = 1/16$
 - $P_X(X = 1) = 1/4$
 - $P_X(X = 2) = 3/8$
 - $P_X(X = 3) = 1/4$
 - $P_X(X = 4) = 1/16$

With every random variable X , we associate a function called the cumulative distribution function of X .

Definition 13

The **cumulative distribution function** or cdf of a random variable X , denoted by $F_X(x)$, is defined by

$$F_X(x) = P_X(X \leq x), \quad \text{for all } x.$$

Theorem 14

The function $F(x)$ is a cdf if and only if the following three conditions hold:

- ❶ $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.
- ❷ $F(x)$ is a nondecreasing function of x .
- ❸ $F(x)$ is right-continuous; that is, for every number x_0 , $\lim_{x \downarrow x_0} F(x) = F(x_0)$.

Definition 15

A random variable X is **continuous** if $F_X(x)$ is a continuous function of x . A random variable X is **discrete** if $F_X(x)$ is a step function of x . There are random variables which are neither continuous nor discrete. For example, they can be continuous over some range and discrete in another range.

Remark:

A distribution function F_X completely characterizes the probability distribution of a random variable X .

Definition 16

The random variables X and Y are **identically distributed** if, for every set $A \in \mathcal{B}$, $P_X(X \in A) = P_Y(Y \in A)$.

Theorem 17

The following two statements are equivalent:

- 1 The random variables X and Y are identically distributed.
- 2 $F_X(x) = F_Y(x)$ for every x .

Density and Mass functions

Definition 18

The **probability mass function** (pmf) of a discrete random variable X is given by

$$f_X(x) = P_X(X = x)$$

Definition 19

The **probability density function** (pdf), $f_X(x)$, of a *continuous* random variable X is the function that satisfies

$$F_X(x) = \int_{-\infty}^x f_X(t) dt \quad \text{for all } x.$$

Theorem 20

A function $f_X(x)$ is a pdf (or pmf) of a random variable X if and only if

- ① $f_X(x) \geq 0$ for all x .
- ② $\sum_x f_X(x) = 1$ (pmf) or $\int_{-\infty}^{\infty} f_X(x) = 1$ (pdf).

- We say that X is (absolutely) continuous RV if there exists a non-negative function f , defined for all $x \in (-\infty, \infty)$ having the property that for any (measurable) set B of real numbers,

$$\Pr(X \in B) = \int_B f(x)dx$$

- As we saw before f is called the **probability density function** of X .
- All probability statements about X can be answered in terms of f .
- If $B = [a, b]$, we obtain

$$\Pr(a \leq X \leq b) = \int_a^b f(x)dx$$

- The relationship between the distribution function F and the probability density f is expressed by:

$$F(a) = \Pr(X \in (-\infty, a]) = \int_{-\infty}^a f(x)dx$$

- Using the Fundamental Theorem of Calculus, differentiating both sides we obtain:

$$\frac{d}{da}F(a) = f(a)$$

- Analogously to the discrete case:

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

Proposition 21

If X is a continuous random variable with probability density function $f(x)$, then, for any real-valued function g

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

Lemma 22

For a nonnegative random variable Y ,

$$E(Y) = \int_0^{\infty} \Pr(Y > y)dy$$

Proposition 23

If a and b are constants, then

$$E[aX + b] = aE[X] + b$$

- The variance of a continuous random variable X with mean μ , is defined as

$$\text{Var}(X) = E[(X - \mu)^2]$$

- The alternative formula:

$$\text{Var}(X) = E[X^2] - (E[X])^2 = E[X^2] - \mu^2$$

Jointly Distributed Random Variables

- For any two random variables X and Y , the **joint cumulative probability distribution function** of X and Y is

$$F(a, b) = \Pr(X \leq a, Y \leq b), \quad -\infty < a, b < \infty$$

- The distribution of X can be obtained from the joint distribution of X and Y as follows

$$\begin{aligned} F_X(a) &= \Pr(X \leq a) \\ &= \Pr(X \leq a, Y \leq \infty) \\ &= \Pr\left(\lim_{b \rightarrow \infty} \{X \leq a, Y \leq b\}\right) &= \lim_{b \rightarrow \infty} \Pr(X \leq a, Y \leq b) \\ &= \lim_{b \rightarrow \infty} F(a, b) \\ &= F(a, \infty) \end{aligned}$$

- Similarly for $F_Y(b)$. F_X and F_Y are called the **marginal distributions of X and Y respectively**.

- In principle, all joint statements about X and Y can be answered in terms of their joint distribution function.

- $\Pr(X > a, Y > b)$

- $\Pr(a_1 < X \leq a_2, b_1 < Y \leq b_2)$ whenever $a_1 < a_2, b_1 < b_2$.

Jointly continuous random variables

- We say that X and Y are **jointly continuous** if there exists a function $f(x, y)$, defined for all real x and y , having the property that, for every (measurable) set C of pairs of real numbers

$$\Pr(X, Y) \in C = \iint_{(x,y) \in C} f(x, y) dx dy$$

- $f(x, y)$ is called the **joint probability density function** of X and Y .
- if A and B are any sets of (measurable) real numbers, then, defining $C = A \times B$, we get

$$\Pr(X \in A, Y \in B) = \int_B \int_A f(x, y) dx dy$$

- Because

$$F(a, b) = \Pr(X \in (-\infty, a], Y \in (-\infty, b]) = \int_{-\infty}^b \int_{-\infty}^a f(x, y) dx dy$$

it follows that

$$f(a, b) = \frac{\partial^2}{\partial a \partial b} F(a, b)$$

If X and Y are jointly continuous, they are individually continuous, and their probability density functions can be obtained as follows:

$$\Pr(X \in A) = \Pr(X \in A, Y \in (-\infty, \infty)) = \int_A \int_{-\infty}^{\infty} f(x, y) dy dx = \int_A f_X(x) dx$$

where $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$. Analogously for $f_Y(y)$

Independent Random Variables

- The random variables X and Y are said to be **independent** if, for any two sets of real numbers A and B ,

$$\Pr(X \in A, Y \in B) = \Pr(X \in A) \Pr(Y \in B)$$

- In terms of the joint distribution F , X and Y are independent if

$$F(a, b) = F_X(a)F_Y(b), \quad \text{for all } a \text{ and } b.$$

- In the jointly continuous case, the condition of independence is equivalent to

$$f(x, y) = f_X(x)f_Y(y)$$

Proposition 24

The continuous (discrete) random variables X and Y are independent if and only if their joint probability density (mass) function can be expressed as

$$f_{X,Y}(x, y) = h(x)g(y) \quad -\infty < x < \infty, -\infty < y < \infty$$